
A1 interface: Use Cases and Requirements

Copyright © 2025 by the O-RAN ALLIANCE e.V.

The copying or incorporation into any other work of part or all of the material available in this specification in any form without the prior written permission of O-RAN ALLIANCE e.V. is prohibited, save that you may print or download extracts of the material of this specification for your personal use, or copy the material of this specification for the purpose of sending to individual third parties for their information provided that you acknowledge O-RAN ALLIANCE as the source of the material and that you inform the third party that these conditions apply to them and that they must comply with them.

O-RAN ALLIANCE e.V., Buschkauler Weg 27, 53347 Alfter, Germany

Register of Associations, Bonn VR 11238, VAT ID DE321720189

Contents

Foreword.....	5
Modal verbs terminology.....	5
1 Scope	6
2 References	6
2.1 Normative references	6
2.2 Informative references.....	6
3 Definition of terms, symbols, abbreviations and conventions.....	7
3.1 Terms.....	7
3.2 Symbols.....	7
3.3 Abbreviations	7
3.4 Conventions.....	7
3.4.1 General.....	7
3.4.2 Requirements	7
3.4.3 UML diagrams.....	7
4 General	8
5 Requirements.....	8
5.1 Functional requirements	8
6 Use cases for A1 policy management	10
6.1 Policy type discovery use cases.....	10
6.1.1 Background and goal of the use cases	10
6.1.2 Entities/resources involved in the use cases.....	10
6.1.3 Solutions	10
6.1.4 Required data	13
6.2 Policy type status use cases	13
6.2.1 Background and goal of the use cases	13
6.2.2 Entities/resources involved in the use cases.....	13
6.2.3 Solutions	14
6.2.4 Required data	17
6.3 Create policy use cases.....	17
6.3.1 Background and goal of the use cases	17
6.3.2 Entities/resources involved in the use cases.....	17
6.3.3 Solutions	18
6.3.4 Required data	18
6.4 Query policy use cases	18
6.4.1 Background and goal of the use cases	18
6.4.2 Entities/resources involved in the use cases.....	18
6.4.3 Solutions	19
6.4.4 Required data	20
6.5 Update policy use cases.....	20
6.5.1 Background and goal of the use cases	20
6.5.2 Entities/resources involved in the use cases.....	20
6.5.3 Solutions	21
6.5.4 Required data	21
6.6 Delete policy use cases.....	21
6.6.1 Background and goal of the use cases	21
6.6.2 Entities/resources involved in the use cases.....	22
6.6.3 Solutions	22
6.6.4 Required data	22
6.7 Policy status use cases.....	22
6.7.1 Background and goal of the use cases	22
6.7.2 Entities/resources involved in the use cases.....	23
6.7.3 Solutions	23

6.7.4	Required data	24
7	Use cases for A1 enrichment information	24
7.1	EI discovery use cases	24
7.1.1	Background and goal of the use cases	24
7.1.2	Entities/resources involved in the use cases.....	25
7.1.3	Solutions	25
7.2	EI type status use cases	26
7.2.1	Background and goal of the use cases	26
7.2.2	Entities/resources involved in the use cases.....	27
7.2.3	Solutions	27
7.2.4	Required data	30
7.3	Create EI job use cases	30
7.3.1	Background and goal of the use cases	30
7.3.2	Entities/resources involved in the use cases.....	30
7.3.3	Solutions	31
7.3.4	Required data	31
7.4	Query EI jobs use cases.....	31
7.4.1	Background and goal of the use cases	31
7.4.2	Entities/resources involved in the use cases.....	31
7.4.3	Solutions	32
7.4.4	Required data	33
7.5	Update EI job use cases.....	33
7.5.1	Background and goal of the use cases	33
7.5.2	Entities/resources involved in the use cases.....	33
7.5.3	Solutions	34
7.5.4	Required data	34
7.6	Delete EI job use cases	34
7.6.1	Background and goal of the use cases	34
7.6.2	Entities/resources involved in the use cases.....	35
7.6.3	Solutions	35
7.6.4	Required data	35
7.7	EI job status use cases	36
7.7.1	Background and goal of the use cases	36
7.7.2	Entities/resources involved in the use cases.....	36
7.7.3	Solutions	36
7.7.4	Required data	37
7.8	EI delivery use cases	38
7.8.1	Background and goal of the use cases	38
7.8.2	Entities/resources involved in the use cases.....	38
7.8.3	Solutions	38
7.8.4	Required data	39
8	Use cases for A1 ML model management service – Near-RT RIC as A1-ML Consumer	39
8.1	Request AI/ML model training use cases	39
8.1.1	Background and goal of the use cases	39
8.1.2	Entities/resources involved in the use cases.....	39
8.1.3	Solutions	40
8.1.4	Required data	41
8.2	AI/ML model training job status use cases	41
8.2.1	Background and goal of the use cases	41
8.2.2	Entities/resources involved in the use cases.....	41
8.2.3	Solutions	42
8.2.4	Required data	43
8.3	Discover AI/ML model training capabilities use cases	43
8.3.1	Background and goal of the use cases	43
8.3.2	Entities/resources involved in the use cases.....	43
8.3.3	Solutions	44
8.3.4	Required data	44
9	Use cases for A1 ML model management service – Non-RT RIC as A1-ML Consumer	45

9.1	Request AI/ML model training use cases	45
9.1.1	Background and goal of the use cases	45
9.1.2	Entities/resources involved in the use cases.....	45
9.1.3	Solutions	45
9.1.4	Required data	46
9.2	AI/ML model training job status use cases	46
9.2.1	Background and goal of the use cases	46
9.2.2	Entities/resources involved in the use cases.....	46
9.2.3	Solutions	47
9.2.4	Required data	48
Annex A (normative): Authorization of service access requests		49
A.1	Pre-conditions.....	49
A.2	A1 service access based on token verification use cases	49
A.2.1	Background and goal of the use cases	49
A.2.2	Entities/resources involved in the use cases.....	49
A.2.3	Solutions	49
A.2.4	Required data	50
Annex (informative): Change History		51

Foreword

This Technical Specification (TS) has been produced by O-RAN Alliance Working Group 2. It is part of a TS-family covering the A1 interface as identified below:

- "A1 interface: General Aspects and Principles";
- "A1 interface: Use Cases and Requirements";
- "A1 interface: Transport Protocol";
- "A1 interface: Application Protocol";
- "A1 interface: Type Definitions"; and
- "A1 interface: Test Specification".

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN Alliance modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

version xx.yy.zz

where:

- xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.
- yy: the second digit-group is incremented when editorial only changes have been incorporated in the document. Always 2 digits with leading zero if needed.
- zz: the third digit-group included only in working versions of the document indicating incremental changes during the editing process. External versions never include the third digit-group. Always 2 digits with leading zero if needed.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1 Scope

The present document specifies the use cases and requirements for the A1 interface.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

- [1] O-RAN.WG2.TS.A1GAP: "A1 interface: General Aspects and Principles" ("A1GAP").
- [2] Void
- [3] O-RAN.WG2.TS.A1AP: "A1 interface: Application Protocol" ("A1AP").
- [4] O-RAN.WG2.TS.A1TD: "A1 interface: Type Definitions" ("A1TD").
- [5] O-RAN.WG2.TS.Non-RT-RIC-ARCH: "Non-RT RIC Architecture".
- [6] O-RAN.WG3.TS.RICARCH: "Near-RT RIC Architecture".
- [7] Recommendation ITU-T M.3020: "Management interface specification methodology".
- [8] OMG® Unified Modeling Language (OMG UML) version 2.5.1 ("OMG UML").
- [9] O-RAN.WG11.TS.SecProtSpec.0: "O-RAN Security Protocols Specifications" ("SPS").

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

Not applicable.

3 Definition of terms, symbols, abbreviations and conventions

3.1 Terms

For the purposes of the present document, the terms given in A1GAP [1], clause 3.1 apply.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in A1GAP [1], clause 3.3 apply.

3.4 Conventions

3.4.1 General

For the purposes of the present document, the conventions in the following clauses apply.

3.4.2 Requirements

The requirements and solutions for use cases are based on the methodology specified in ITU-T M.3020 [7], clause A.1.2.

3.4.3 UML diagrams

The sequence diagrams for the messages that are exchanged between use case actors are based on the OMG UML [8], clause 17.4.4.1 and an example is illustrated in figure 3.4.3-1.

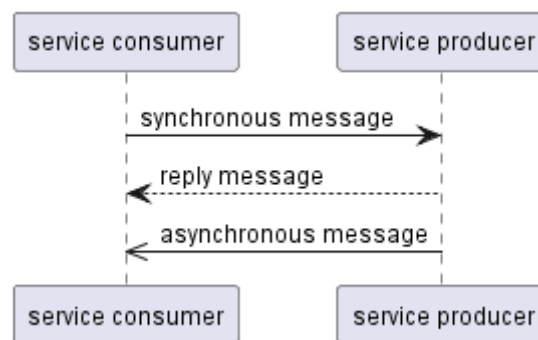


Figure 3.4.3-1: Example of UML messages.

For each message in the present document, the following information is indicated:

- interface name and procedure name;
- type of message (e.g. request or response); and
- transferred information (if applicable).

An example is illustrated in figure 3.4.3-2.

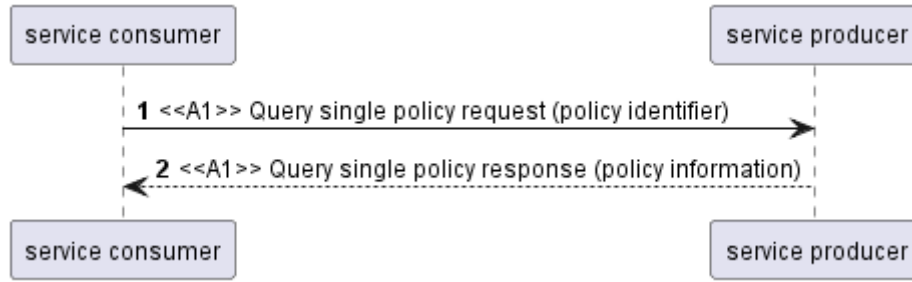


Figure 3.4.3-2: Example of message labels used in the present document.

4 General

Non-RT RIC with A1 termination function and A1 related functions in the SMO/Non-RT RIC framework is defined in Non-RT RIC Architecture [5].

Near-RT RIC with A1 Termination and internal functions is defined in Near-RT RIC Architecture [6].

Access token request, and service access request based on token verification, is defined in SPS [9], clause 4.7.

The A1 policy management service (A1-P) and the A1-P procedures are defined in A1GAP [1], clause 4.1.3.1, the service operations and the API are defined in A1AP [3], clause 6.2, and policy types are defined in A1TD [4], clause 7.2.

The A1 enrichment information service (A1-EI) and the A1-EI procedures are defined in A1GAP [1], clause 4.1.3.2, the service operations and the API are defined in A1AP [3], clause 6.3, and EI types are defined in A1TD [4], clause 9.2.

The A1 ML model management service (A1-ML) is defined in A1GAP [1], clause 4.1.3.3.

5 Requirements

5.1 Functional requirements

For the A1 policy management service (A1-P), the functional requirements are included in table 5.1-1.

Table 5.1-1: A1 Policy management service (A1-P) functional requirements

REQ	Description	Note
REQ-A1-P-FUN1	A1-P shall support discovery of policy type identifiers	
REQ-A1-P-FUN2	An A1 policy type shall be identified by a policy type identifier	
REQ-A1-P-FUN3	A1-P shall support retrieval of information about A1 policy types	
REQ-A1-P-FUN4	A1-P shall support retrieval of status information related to an A1 policy type	
REQ-A1-P-FUN5	A1-P shall support subscription to, and notification about, changes in status information related to an A1 policy type	
REQ-A1-P-FUN6	A1-P shall support request of policy enforcement related to an A1 policy type by managing an A1 policy	
REQ-A1-P-FUN7	A1-P shall support discovery of policy identifiers	
REQ-A1-P-FUN8	A1-P shall support retrieval of information about an A1 policy	
REQ-A1-P-FUN9	A1-P shall support retrieval of status information related to an A1 policy	
REQ-A1-P-FUN10	A1-P shall support subscription to, and notification about, changes in status information related to an A1 policy	
REQ-A1-P-FUN11	A1-P shall support authorization of service access requests	

For the A1 enrichment information service (A1-EI), the functional requirements are included in table 5.1-2.

Table 5.1-2: A1 enrichment information service (A1-EI) functional requirements

REQ	Description	Note
REQ-A1-EI-FUN1	A1-EI shall support discovery of EI type identifiers	
REQ-A1-EI-FUN2	An EI type shall be identified by an EI type identifier	
REQ-A1-EI-FUN3	A1-EI shall support retrieval of information about EI types	
REQ-A1-EI-FUN4	A1-EI shall support retrieval of status information related to an EI type	
REQ-A1-EI-FUN5	A1-EI shall support subscription to, and notification about, changes in status information related to an EI type	
REQ-A1-EI-FUN6	A1-EI shall support request of enrichment information related to an EI type by managing an EI job	
REQ-A1-EI-FUN7	A1-EI shall support discovery of EI job identifiers	
REQ-A1-EI-FUN8	A1-EI shall support retrieval of information about an EI job	
REQ-A1-EI-FUN9	A1-EI shall support retrieval of status information related to an EI job	
REQ-A1-EI-FUN10	A1-EI shall support subscription to, and notification about, changes in status information related to an EI job.	
REQ-A1-EI-FUN11	A1-EI shall support delivery of enrichment information over the A1 interface	
REQ-A1-EI-FUN12	A1-EI shall support authorization of service access requests	

For the A1 ML model management service (A1-ML), the functional requirements are included in table 5.1-3.

Table 5.1-3: A1 ML model management service (A1-ML) functional requirements

REQ	Description	Note
REQ-A1-ML-FUN1	A1-ML shall support authorization of service access requests	
REQ-A1-ML-FUN2	A1-ML shall support the request of AI/ML model training	
REQ-A1-ML-FUN3	A1-ML shall support query of AI/ML model training job status.	
REQ-A1-ML-FUN4	A1-ML shall support notification for AI/ML model training job status changes.	
REQ-A1-ML-FUN5	A1-ML shall support discovery of AI/ML model training capabilities.	

6 Use cases for A1 policy management

6.1 Policy type discovery use cases

6.1.1 Background and goal of the use cases

The policy type discovery use cases define how Non-RT RIC can detect which A1 policy types are available in a Near-RT RIC and how Non-RT RIC can retrieve information about one or more A1 policy types.

Policy type information is provided by the Near-RT RIC and is used by Non-RT RIC for creating and by Near-RT RIC for validating A1 policies. When policy type identifier and policy type information are known, the Non-RT RIC can create A1 policies in a Near-RT RIC as described by policy creation use cases.

6.1.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:
 - a) Discovers A1 policy types available in Near-RT RIC.
- 2) Near-RT RIC:
 - b) Makes A1 policy types available to Non-RT RIC for which it can support A1 policies.

6.1.3 Solutions

6.1.3.1 Query all policy type identifiers

Table 6.1.3.1-1: Query all policy type identifiers use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to discover policy type identifiers for A1 policy types that are available in Near-RT RIC	
Actors and Roles	Non-RT RIC as A1-P Consumer Near-RT RIC as A1-P Producer	
Assumptions	A1 policy types are available in Near-RT RIC	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-P service	
Begins when	Non-RT RIC initiates A1 policy type identifiers query	
Step 1 (M)	Non-RT RIC sends Query policy type identifiers request	
Step 2 (M)	Near-RT RIC sends Query policy type identifiers response containing policy type identifiers	
Ends when	Non-RT RIC has received policy type identifiers for all available A1 policy types	
Exceptions		
Post-conditions	Policy type identifiers are known to Non-RT RIC	
Traceability	REQ-A1-P-FUN1	

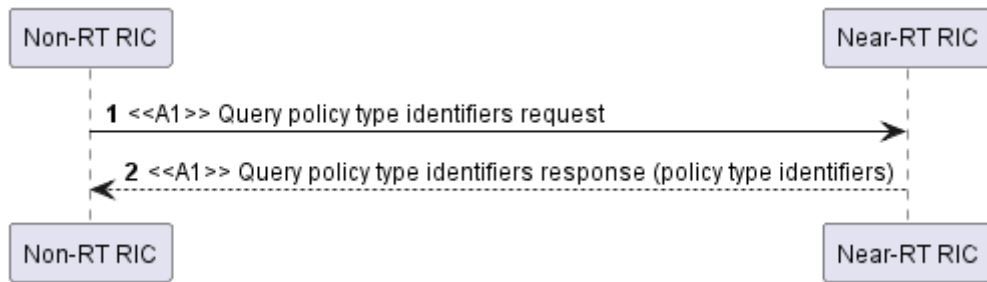


Figure 6.1.3.1-1: Query all policy type identifiers

6.1.3.2 Query single policy type

Table 6.1.3.2-1: Query single policy type use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to retrieve policy type information about an A1 policy type	
Actors and Roles	Non-RT RIC as A1-P Consumer Near-RT RIC as A1-P Producer	
Assumptions	The policy type identifier known to the Non-RT RIC corresponds to an available A1 policy type	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-P service. The policy type identifier is known to the Non-RT RIC.	
Begins when	Non-RT RIC initiates A1 policy type query	
Step 1 (M)	Non-RT RIC sends Query single policy type request containing the policy type identifier of the policy type being queried	
Step 2 (M)	Near-RT RIC sends Query single policy type response containing policy type information	
Ends when	Non-RT RIC has received the A1 policy type information	
Exceptions		
Post-conditions	A1 policy type information is known to Non-RT RIC	
Traceability	REQ-A1-P-FUN2, REQ-A1-P-FUN3	

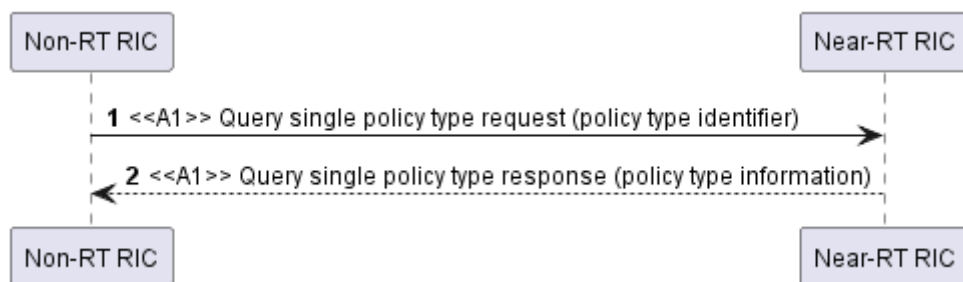


Figure 6.1.3.2-1: Query single policy type

6.1.3.3 Query multiple policy types

Table 6.1.3.3-1: Query multiple policy types use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to retrieve policy type information for a selection of A1 policy types it is interested in	
Actors and Roles	Non-RT RIC as A1-P Consumer Near-RT RIC as A1-P Producer	
Assumptions	The policy type identifiers known to the Non-RT RIC correspond to available A1 policy types	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-P service. The policy type identifiers are known to the Non-RT RIC.	
Begins when	Non-RT RIC initiates A1 policy types query	
Step 1 - ref (M)	Non-RT RIC queries for information about a single policy type	6.1.3.2
Step 2 - loop (M)	Non-RT RIC repeats Step 1 for the policy type identifiers it is interested in	
Ends when	Non-RT RIC has received information about multiple available A1 policy types	
Exceptions		
Post-conditions	Information about multiple A1 policy types is known to Non-RT RIC	
Traceability	REQ-A1-P-FUN2, REQ-A1-P-FUN3	

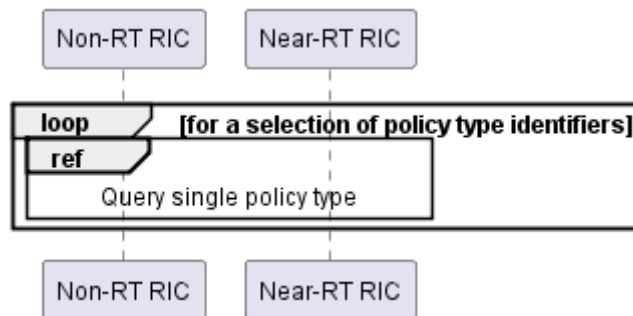


Figure 6.1.3.3-1: Query multiple policy types

6.1.3.4 Query all policy type information

Table 6.1.3.4-1: Query all policy type information use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to retrieve policy type information for all A1 policy types that are available in Near-RT RIC	
Actors and Roles	Non-RT RIC as A1-P Consumer Near-RT RIC as A1-P Producer	
Assumptions	The policy type identifiers known to the Non-RT RIC correspond to all available A1 policy types.	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-P service. All policy type identifiers are known to the Non-RT RIC.	
Begins when	Non-RT RIC initiates A1 policy types query	
Step 1 - ref (M)	Non-RT RIC queries for information about a single policy type	6.1.3.2
Step 2 - loop (M)	Non-RT RIC repeats Step 1 for all policy type identifiers	
Ends when	Non-RT RIC has received information about all available A1 policy types	
Exceptions		
Post-conditions	Information about all available A1 policy types is known to Non-RT RIC	
Traceability	REQ-A1-P-FUN2, REQ-A1-P-FUN3	

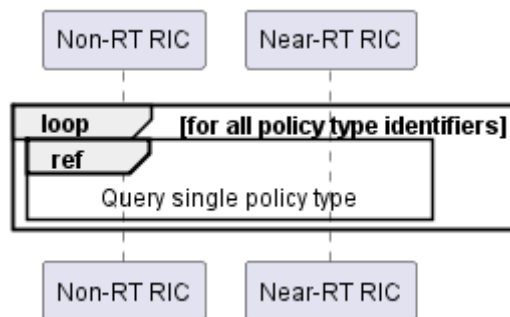


Figure 6.1.3.4-1: Query all policy types

6.1.4 Required data

An A1 policy type is identified by a policy type identifier. A list of policy type identifiers is provided by Near-RT RIC corresponding to available A1 policy types. The Non-RT RIC provides a policy type identifier when querying for information about a single A1 policy type. The Near-RT RIC includes policy type information in response to a query for information about an A1 policy type.

6.2 Policy type status use cases

6.2.1 Background and goal of the use cases

The policy type status use cases define how Non-RT RIC can detect status of an A1 policy type and subscribe to notifications for changes in availability and state of policy types.

Policy type status information is provided by the Near-RT RIC and is used by Non-RT RIC when managing A1 policies.

6.2.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:

- a) Requests policy type status.
- b) Subscribes to notifications for changes in policy type status.
- 2) Near-RT RIC:
 - a) Responds to queries for policy type status.
 - b) Handles subscriptions and notifies Non-RT RIC about changes in availability and state of policy types.

6.2.3 Solutions

6.2.3.1 Query policy type status

Table 6.2.3.1-1: Query policy type status use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to retrieve policy type status information for an A1 policy type	
Actors and Roles	Non-RT RIC as A1-P Consumer Near-RT RIC as A1-P Producer	
Assumptions	The policy type identifier known to the Non-RT RIC corresponds to an available policy type	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-P service. The policy type identifier is known to the Non-RT RIC.	
Begins when	Non-RT RIC initiates A1 policy type status query	
Step 1 (M)	Non-RT RIC sends Query policy type status request containing the policy type identifier of the policy type being queried for status	
Step 2 (M)	Near-RT RIC sends Query policy type status response containing policy type status information	
Ends when	Non-RT RIC has received the A1 policy type status information	
Exceptions		
Post-conditions	Policy type status information is known to Non-RT RIC	
Traceability	REQ-A1-P-FUN4	

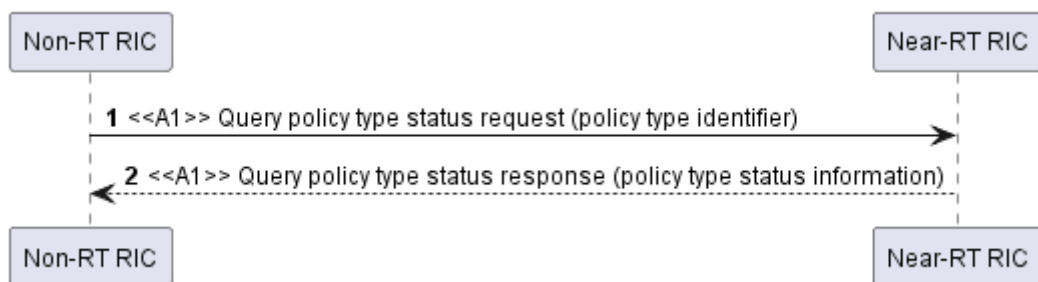


Figure 6.2.3.3-1: Query policy type status

6.2.3.2 Subscription for policy type status notification

Table 6.2.3.2-1: Subscribe policy type status use case

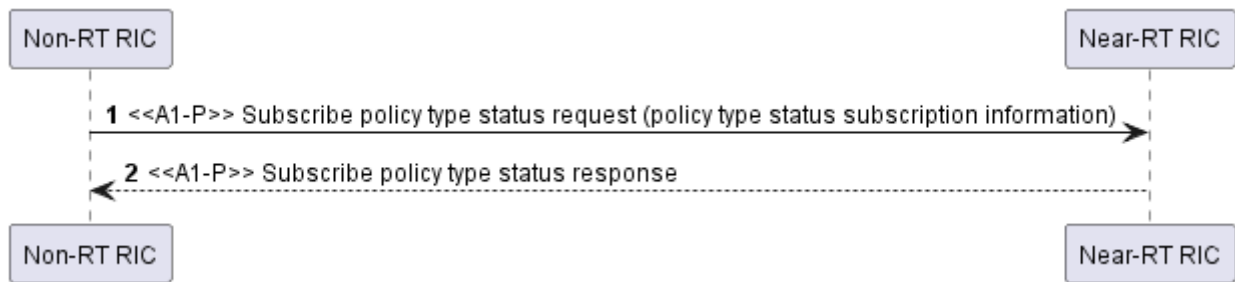
Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to subscribe to notifications for policy type status information for A1 policy types	
Actors and Roles	Non-RT RIC as A1-P Consumer Near-RT RIC as A1-P Producer	
Assumptions	Non-RT RIC is interested in notifications of policy types that are made available or unavailable, and/or notifications of state changes of a policy type	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-P service. Policy type identifiers are known to the Non-RT RIC when subscribing for notifications on type state changes.	
Begins when	Non-RT RIC initiates A1 Policy type status subscribe	
Step 1 (M)	Non-RT RIC sends Subscribe policy type status request containing the policy type status subscription information including policy type identifier(s) if type state changes are requested.	
Step 2 (M)	Near-RT RIC sends Subscribe policy type status response	
Ends when	Policy type status subscription has been created	
Exceptions		
Post-conditions	Non-RT RIC is subscribed to policy type status notifications	
Traceability	REQ-A1-P-FUN5	

Table 6.2.3.2-2: Update policy type status subscription use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to update its subscription to notifications for policy type status information for A1 policy types	
Actors and Roles	Non-RT RIC as A1-P Consumer Near-RT RIC as A1-P Producer	
Assumptions	Non-RT RIC is interested in notifications of policy types that are made available or unavailable, and/or notifications of state changes of a policy type	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-P service. Policy type status notifications has been subscribed to.	
Begins when	Non-RT RIC initiates A1 Policy type status subscribe	
Step 1 (M)	Non-RT RIC sends Subscribe policy type status request containing the updated policy type status subscription information	
Step 2 (M)	Near-RT RIC sends Subscribe policy type status response	
Ends when	Policy type status subscription has been updated	
Exceptions		
Post-conditions	Non-RT RIC is subscribed to policy type status notifications	
Traceability	REQ-A1-P-FUN5	

Table 6.2.3.2-3: Unsubscribe policy type status use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to unsubscribe from notifications for policy type status information for A1 policy types	
Actors and Roles	Non-RT RIC as A1-P Consumer Near-RT RIC as A1-P Producer	
Assumptions	Non-RT RIC is no longer interested in notifications of policy types status	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-P service. Policy type status notifications has been subscribed to.	
Begins when	Non-RT RIC initiates A1 Policy type status unsubscribe	
Step 1 (M)	Non-RT RIC sends Subscribe policy type status request containing policy type status subscription information with no content	
Step 2 (M)	Near-RT RIC sends Subscribe policy type status response	
Ends when	Policy type status subscription has been deleted	
Exceptions		
Post-conditions	Non-RT RIC is not subscribed to policy type status notifications	
Traceability	REQ-A1-P-FUN5	


Figure 6.2.3.2-1: Subscription for policy type status notifications

6.2.3.3 Notify policy type status

Table 6.2.3.3-1: Notify policy type status use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to receive policy type status information for an A1 Policy type	
Actors and Roles	Non-RT RIC as A1-P Consumer Near-RT RIC as A1-P Producer	
Assumptions		
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-P service. Non-RT RIC has subscribed to policy type status notifications and provided a callback URI.	
Begins when	Event occurs in Near-RT RIC related to the status of an A1 Policy type, either a change in availability or of the type state	
Step 1 (M)	Near-RT RIC sends Notify policy type status message containing the policy type status information	
Ends when	Non-RT RIC has received the policy type status information	
Exceptions		
Post-conditions	Policy type status information is known to Non-RT RIC	
Traceability	REQ-A1-P-FUN5	

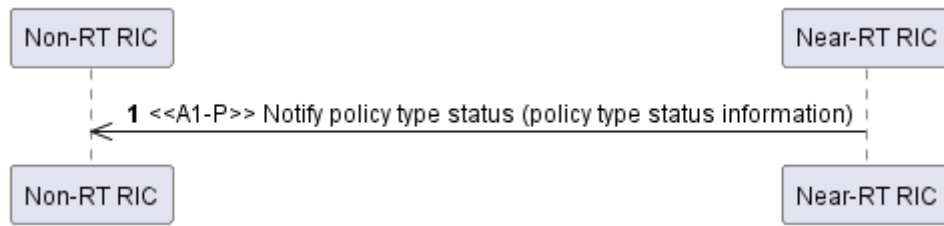


Figure 6.2.3.3-1: Notify policy type status

6.2.4 Required data

The policy type status information includes policy type availability change information (i.e. if A1 policy type has been made available or unavailable) and/or state change information related to an A1 policy type.

The state of the A1 policy type indicates if A1 policies can be created for the policy type or not, and if A1 policies for the policy type would be enforced or not enforced.

The policy type status subscription information includes details on whether notifications are requested for changes in A1 policy type availability and/or changes in state of the A1 policy type for which policy type identifiers were provided.

6.3 Create policy use cases

6.3.1 Background and goal of the use cases

The create policy use cases define how Non-RT RIC can create an A1 policy for an A1 policy type and subscribe to notifications for changes in policy status.

6.3.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:
 - a) Creates A1 policy in Near-RT RIC.
- 2) Near-RT RIC:
 - b) Enforces A1 policies for available A1 policy types.

6.3.3 Solutions

6.3.3.1 Create single policy

Table 6.3.3.1-1: Create single policy use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to create A1 policy for an A1 policy type	
Actors and Roles	Non-RT RIC as A1-P Consumer Near-RT RIC as A1-P Producer	
Assumptions	The policy type information known to the Non-RT RIC corresponds to an available A1 policy type. Non-RT RIC has the schema for formulating A1 policy information.	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-P service. The policy type information is known to the Non-RT RIC.	
Begins when	Non-RT RIC initiates A1 policy creation	
Step 1 (M)	Non-RT RIC sends Create policy request containing the policy identifier and the policy information	
Step 2 (M)	Near-RT RIC sends Create policy response	
Ends when	A1 policy has been created	
Exceptions		
Post-conditions	The A1 policy exists	
Traceability	REQ-A1-P-FUN6	

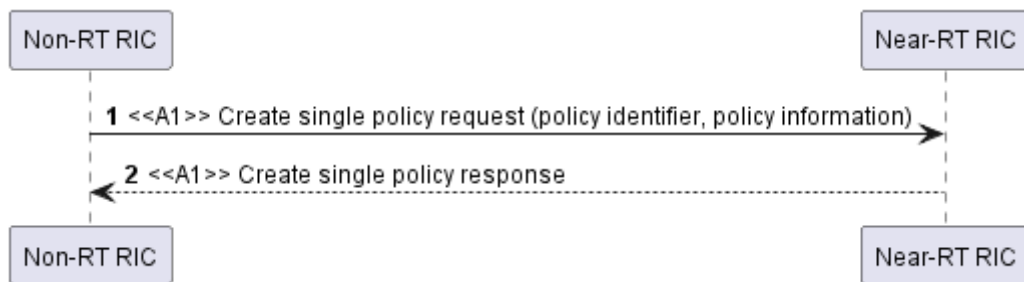


Figure 6.3.3.4-1: Create single policy

6.3.4 Required data

For creating an A1 policy of a certain policy type, the Non-RT RIC provides the policy identifier and the policy information, the callback URI for policy result delivery, and optionally the callback URI for policy status notifications.

6.4 Query policy use cases

6.4.1 Background and goal of the use cases

The query policies use cases define how Non-RT RIC can query for information on existing A1 policies.

6.4.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:
 - a) Discovers A1 policies that exist in the Near-RT RIC.

- b) Retrieves policy information.
- 2) Near-RT RIC:
 - a) Handles A1 policies.
 - b) Responds to queries for policy identifiers and policy information.

6.4.3 Solutions

6.4.3.1 Query policy identifiers

Table 6.4.3.1-1: Query policy identifiers use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to discover policy identifiers for A1 policies that exist in Near-RT RIC	
Actors and Roles	Non-RT RIC as A1-P Consumer Near-RT RIC as A1-P Producer	
Assumptions	A1 policies exist in Near-RT RIC	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-P service	
Begins when	Non-RT RIC initiates A1 policy identifiers query	
Step 1 (M)	Non-RT RIC sends Query policy identifiers request	
Step 2 (M)	Near-RT RIC sends Query policy identifiers response containing policy identifiers	
Ends when	Non-RT RIC has received policy identifiers for existing A1 policies	
Exceptions		
Post-conditions	Policy identifiers are known to Non-RT RIC	
Traceability	REQ-A1-P-FUN7	

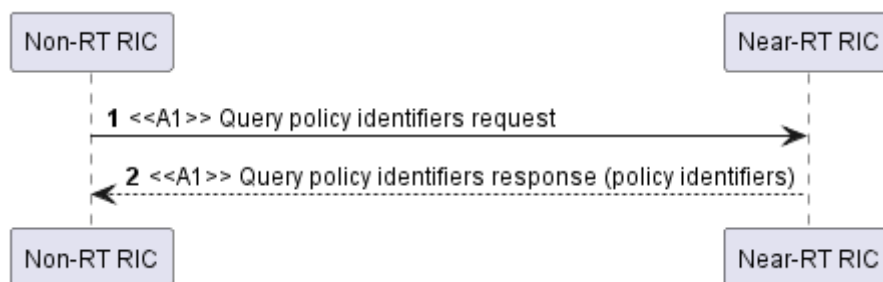


Figure 6.4.3.1-1: Query policy identifiers

6.4.3.2 Query single policy

Table 6.4.3.2-1: Query single policy use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to retrieve information about an A1 policy	
Actors and Roles	Non-RT RIC as A1-P Consumer Near-RT RIC as A1-P Producer	
Assumptions	The A1 policy identifier known to the Non-RT RIC corresponds to an existing A1 policy. Non-RT RIC has the schema for interpreting policy information.	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-P service. The policy identifier is known to the Non-RT RIC.	
Begins when	Non-RT RIC initiates A1 policy query	
Step 1 (M)	Non-RT RIC sends Query policy request containing the policy identifier	
Step 2 (M)	Near-RT RIC sends Query policy response containing policy information	
Ends when	Non-RT RIC has received the policy information about the A1 policy	
Exceptions		
Post-conditions	Policy information is known to Near-RT RIC	
Traceability	REQ-A1-P-FUN8	



Figure 6.4.3.2-1: Query single policy

6.4.4 Required data

For querying policy identifiers, the Non-RT RIC optionally provides a policy type identifier as filter parameter.

For querying an A1 policy, the Non-RT RIC provides a policy identifier.

The policy information includes the A1 policy definition.

6.5 Update policy use cases

6.5.1 Background and goal of the use cases

The update policy use cases define how Non-RT RIC can update an existing A1 policy.

Non-RT RIC provides updated policy information. The reason for the update can be related to state of the A1 policy type and received policy status information.

6.5.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:

- a) Updates A1 policy in Near-RT RIC that it has created.
- 2) Near-RT RIC:
 - b) Handles A1 policies for available policy types.

6.5.3 Solutions

6.5.3.1 Update single policy

Table 6.5.3.1-1: Update single policy use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to update an existing A1 policy	
Actors and Roles	Non-RT RIC as A1-P Consumer Near-RT RIC as A1-P Producer	
Assumptions	The policy identifier known to the Non-RT RIC corresponds to an existing A1 policy that was created by the Non-RT RIC	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-P service. The A1 policy exists.	
Begins when	Non-RT RIC initiates A1 policy update	
Step 1 (M)	Non-RT RIC sends Update policy request containing the updated policy information	
Step 2 (M)	Near-RT RIC sends Update policy response	
Ends when	A1 policy has been updated	
Exceptions		
Post-conditions	The A1 policy exists	
Traceability	REQ-A1-P-FUN6	



Figure 6.5.3.4-1: Update single policy

6.5.4 Required data

For updating an A1 policy, the Non-RT RIC provides the policy identifier and updated policy information and/or callback URI for policy status notifications.

6.6 Delete policy use cases

6.6.1 Background and goal of the use cases

The delete policy use cases define how Non-RT RIC can delete an A1 policy.

The reason for the deletion can be related to the state of the A1 policy type and received policy status information.

6.6.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:
 - a) Deletes A1 policy in Near-RT RIC that it has created.
- 2) Near-RT RIC:
 - b) Handles A1 policies for available policy types.

6.6.3 Solutions

6.6.3.1 Delete single policy

Table 6.6.3.1-1: Delete single policy use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to delete an existing A1 policy	
Actors and Roles	Non-RT RIC as A1-P Consumer Near-RT RIC as A1-P Producer	
Assumptions	The policy identifier known to the Non-RT RIC corresponds to an existing A1 policy that was created by the Non-RT RIC	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-P service. The A1 policy exists.	
Begins when	Non-RT RIC initiates A1 policy deletion	
Step 1 (M)	Non-RT RIC sends Delete policy request containing the policy identifier	
Step 2 (M)	Near-RT RIC sends Delete policy response	
Ends when	A1 policy has been deleted	
Exceptions		
Post-conditions	The A1 policy does not exist	
Traceability	REQ-A1-P-FUN6	



Figure 6.6.3.1-1: Delete single policy

6.6.4 Required data

For deleting an A1 policy, the Non-RT RIC provides the policy identifier.

6.7 Policy status use cases

6.7.1 Background and goal of the use cases

The policy status use cases define how Non-RT RIC can detect status, and changes in status, of an A1 policy.

Policy status information is provided by the Near-RT RIC and is used by Non-RT RIC when managing A1 policies.

6.7.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:
 - a) Requests A1 policy status.
 - b) Receives notifications about changes in policy status.
- 2) Near-RT RIC:
 - a) Handles A1 policy and responds to queries for policy status.
 - b) Notifies Non-RT RIC about changes in policy status.

6.7.3 Solutions

6.7.3.1 Query policy status

Table 6.7.3.1-1: Query policy status use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to retrieve policy status information for an existing A1 policy	
Actors and Roles	Non-RT RIC as A1-P Consumer Near-RT RIC as A1-P Producer	
Assumptions	The policy identifier known to the Non-RT RIC corresponds to an existing A1 policy that was created by the Non-RT RIC. Non-RT RIC has the schema for interpreting policy status information.	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-P service. The policy identifier is known to the Non-RT RIC.	
Begins when	Non-RT RIC initiates A1 policy status query	
Step 1 (M)	Non-RT RIC sends Query policy status request containing the policy identifier of the A1 policy being queried for status	
Step 2 (M)	Near-RT RIC sends Query policy status response containing policy status information	
Ends when	Non-RT RIC has received the A1 policy status information	
Exceptions		
Post-conditions	Policy status information is known to Non-RT RIC	
Traceability	REQ-A1-P-FUN9	



Figure 6.7.3.1-1: Query policy status

6.7.3.2 Notify policy status change

Table 6.7.3.2-1: Notify policy status change use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to receive policy status change information for an existing A1 policy	
Actors and Roles	Non-RT RIC as A1-P Consumer Near-RT RIC as A1-P Producer	
Assumptions	Non-RT RIC has the schema for interpreting policy status information.	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-P service. A callback URI for policy status notifications has been provided when creating and/or updating the A1 policy.	
Begins when	Event occurs in Near-RT RIC related to the status change of the A1 policy	
Step 1 (M)	Near-RT RIC sends Notify policy status message containing the policy status information	
Ends when	Non-RT RIC has received the A1 policy status change information	
Exceptions		
Post-conditions	Policy status information is known to Non-RT RIC	
Traceability	REQ-A1-P-FUN10	



Figure 6.7.3.2-1: Notify policy status change

6.7.4 Required data

The policy status information includes the A1 policy status that is formulated based on, and validated against, the policy status schema and indicates if A1 policy is enforced or not enforced.

The policy status change information includes indication about a change of the A1 policy status.

7 Use cases for A1 enrichment information

7.1 EI discovery use cases

7.1.1 Background and goal of the use cases

The EI discovery use cases define how Near-RT RIC can detect which EI types are available in Non-RT RIC and how Near-RT RIC can retrieve information about EI types.

EI type information is provided by Non-RT RIC and is used by Near-RT RIC for creating, and by Non-RT RIC for validating, EI jobs. EI type information is also used by Non-RT RIC for creating, and by Near-RT RIC for validating, EI job results.

When EI type identifier and EI type information are known, the Near-RT RIC can create EI jobs in a Non-RT RIC as described by EI job creation use cases, and the Non-RT RIC can deliver EI job results to Near-RT RIC as described by EI delivery use cases.

EI job constraints information is provided by Non-RT RIC and used by Near-RT RIC when formulating EI job definitions based on the EI type information.

7.1.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:
 - a) Makes EI types available to Near-RT RIC for which it can support EI jobs and deliver EI job results.
 - b) Provides information on EI types and constraints for EI jobs.
- 2) Near-RT RIC:
 - a) Discovers EI types available in Non-RT RIC.
 - b) Retrieves EI type information.

7.1.3 Solutions

7.1.3.1 Query EI type identifiers

Table 7.1.3.1-1: Query EI type identifiers use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC to discover EI type identifiers for EI types that are available in Non-RT RIC	
Actors and Roles	Non-RT RIC as A1-EI Producer Near-RT RIC as A1-EI Consumer	
Assumptions	EI types are available in Non-RT RIC	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-EI service	
Begins when	Near-RT RIC initiates EI type identifiers query	
Step 1 (M)	Near-RT RIC sends Query EI type identifiers request	
Step 2 (M)	Non-RT RIC sends Query EI type identifiers response containing EI type identifiers	
Ends when	Near-RT RIC has received EI type identifiers for available EI types	
Exceptions		
Post-conditions	EI type identifiers are known to Non-RT RIC	
Traceability	REQ-A1-EI-FUN1	

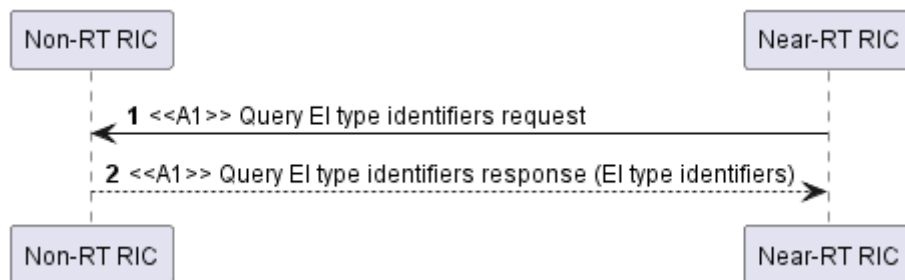


Figure 7.1.3.1-1: Query EI type identifiers

7.1.3.2 Query EI type

Table 7.1.3.2-1: Query EI type use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC to retrieve EI type information about an EI type	
Actors and Roles	Non-RT RIC as A1-EI Producer Near-RT RIC as A1-EI Consumer	
Assumptions	The EI type identifier known to the Near-RT RIC corresponds to an available EI type	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-EI service. The EI type identifier is known to the Near-RT RIC.	
Begins when	Near-RT RIC initiates EI type query	
Step 1 (M)	Near-RT RIC sends Query EI type request containing the EI type identifier of the EI type being queried	
Step 2 (M)	Non-RT RIC sends Query EI type response containing EI type information and EI job constraints information	
Ends when	Near-RT RIC has received the EI type information	
Exceptions		
Post-conditions	EI type information is known to Near-RT RIC	
Traceability	REQ-A1-EI-FUN2, REQ-A1-EI-FUN3	

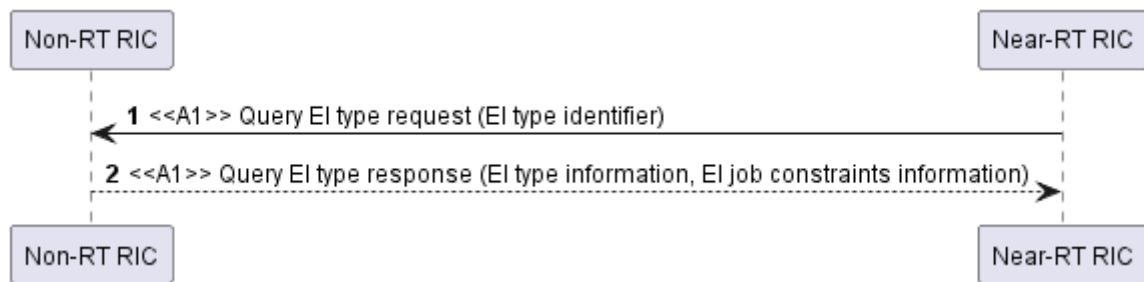


Figure 7.1.3.2-1: Query EI type

7.1.4 Required data

An EI type is identified by an EI type identifier. A list of EI type identifiers is provided by Non-RT RIC corresponding to available EI types. The Near-RT RIC provides an EI type identifier when querying for information about an EI type. In response to a query for information about an EI type, the Non-RT RIC includes EI type information and optionally also EI job constraints information.

The EI type information includes the schemas for EI job definition, EI job status, EI job result and EI job constraints. The EI job constraints information includes information on how EI job can be created and how EI job results can be produced and delivered.

7.2 EI type status use cases

7.2.1 Background and goal of the use cases

The EI type status use cases define how Near-RT RIC can detect status of an EI type and subscribe to notifications for changes in availability and status of EI types.

EI type status information is provided by the Non-RT RIC and is used by Near-RT RIC when managing EI jobs.

7.2.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:
 - a) Responds to queries for EI type status.
 - b) Handles subscriptions and notifies Near-RT RIC about changes in availability and state of EI types.
- 2) Near-RT RIC:
 - a) Requests EI type status.
 - b) Subscribes to notifications for changes in EI type status.

7.2.3 Solutions

7.2.3.1 Query EI type status

Table 7.2.3.1-1: Query EI type status use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC to retrieve EI type status information for an EI type	
Actors and Roles	Non-RT RIC as A1-EI Producer Near-RT RIC as A1-EI Consumer	
Assumptions	The EI type identifier known to the Near-RT RIC corresponds to an available EI type	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-EI service. The EI type identifier is known to the Near-RT RIC.	
Begins when	Near-RT RIC initiates EI type status query	
Step 1 (M)	Near-RT RIC sends Query EI type status request containing the EI type identifier of the EI type being queried for status	
Step 2 (M)	Non-RT RIC sends Query EI type status response containing EI type status information	
Ends when	Near-RT RIC has received the EI type status information	
Exceptions		
Post-conditions	EI type status information is known to Near-RT RIC	
Traceability	REQ-A1-EI-FUN4	

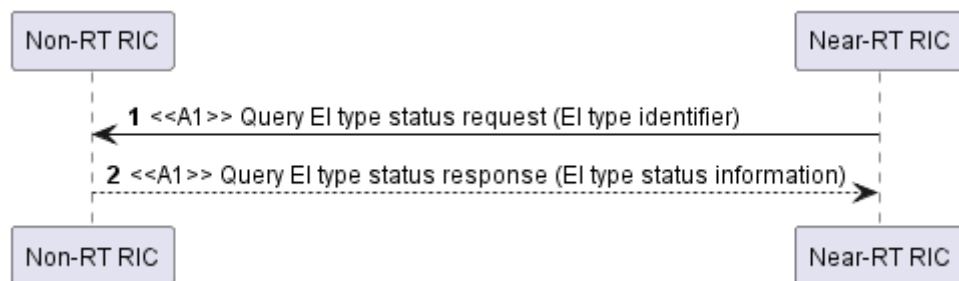


Figure 7.2.3.1-1: Query EI type status

7.2.3.2 Subscription for EI type status notification

Table 7.2.3.2-1: Subscribe EI type status use case

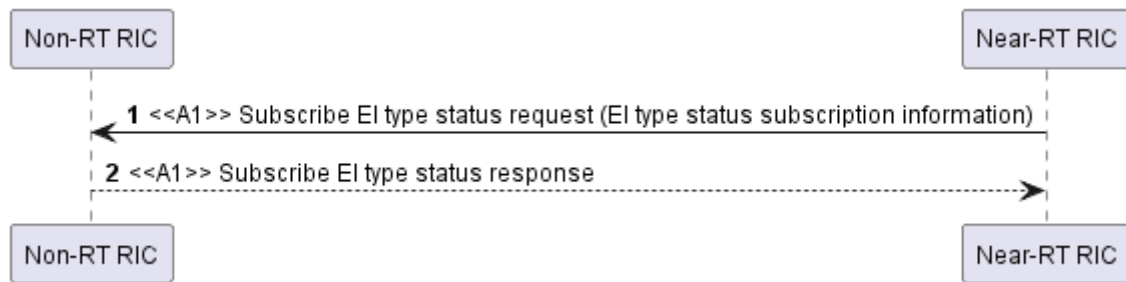
Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC to subscribe to notifications for EI type status information	
Actors and Roles	Non-RT RIC as A1-EI Producer Near-RT RIC as A1-EI Consumer	
Assumptions	Near-RT RIC is interested in notifications of EI types that are made available or unavailable, and/or notifications of state changes of a policy type	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-EI service. EI type identifiers are known to the Near-RT RIC when subscribing for notifications on type state changes.	
Begins when	Near-RT RIC initiates EI type status subscribe	
Step 1 (M)	Near-RT RIC sends Subscribe EI type status request containing the EI type status subscription information including EI type identifier(s) if type state changes are requested.	
Step 2 (M)	Non-RT RIC sends Subscribe EI type status response	
Ends when	EI type status subscription has been created	
Exceptions		
Post-conditions	Near-RT RIC is subscribed to EI type status notifications	
Traceability	REQ-A1-EI-FUN5	

Table 7.2.3.2-2: Update type status subscription use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC to update its subscription to notifications for EI type status information	
Actors and Roles	Non-RT RIC as A1-EI Producer Near-RT RIC as A1-EI Consumer	
Assumptions	Near-RT RIC is interested in notifications of EI types that are made available or unavailable, and/or notifications of state changes of a policy type	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-EI service. EI type status notifications has been subscribed to.	
Begins when	Near-RT RIC initiates EI type status subscribe	
Step 1 (M)	Near-RT RIC sends Subscribe EI type status request containing the updated EI type status subscription information	
Step 2 (M)	Non-RT RIC sends Subscribe EI type status response	
Ends when	EI type status subscription has been updated	
Exceptions		
Post-conditions	Near-RT RIC is subscribed to EI type status notifications	
Traceability	REQ-A1-EI-FUN5	

Table 7.2.3.2-3: Unsubscribe EI type status use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC to unsubscribe from notifications for EI type status information	
Actors and Roles	Non-RT RIC as A1-EI Producer Near-RT RIC as A1-EI Consumer	
Assumptions	Near-RT RIC is no longer interested in notifications of EI types status	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-EI service. EI type status notifications have been subscribed to.	
Begins when	Near-RT RIC initiates EI type status unsubscribe	
Step 1 (M)	Near-RT RIC sends Subscribe EI type status request containing the EI type status subscription information with no content	
Step 2 (M)	Non-RT RIC sends Subscribe EI type status response	
Ends when	EI type status subscription has been deleted	
Exceptions		
Post-conditions	Near-RT RIC is not subscribed to EI type status notifications	
Traceability	REQ-A1-EI-FUN5	


Figure 7.2.3.2-1: Subscribe EI type status

7.2.3.3 Notify EI type status

Table 7.2.3.3-1: Notify EI type status use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC to receive EI type status information for an EI type	
Actors and Roles	Non-RT RIC as A1-EI Producer Near-RT RIC as A1-EI Consumer	
Assumptions		
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-EI service. Near-RT RIC has subscribed to EI type status notifications and provided a callback URI.	
Begins when	Event occurs in Non-RT RIC related to the status of an EI type, either a change in availability or of the type state	
Step 1 (M)	Non-RT RIC sends Notify EI type status request containing the EI type status information	
Ends when	Near-RT RIC has received the EI type status information	
Exceptions		
Post-conditions	EI type status information is known to Near-RT RIC	
Traceability	REQ-A1-EI-FUN5	

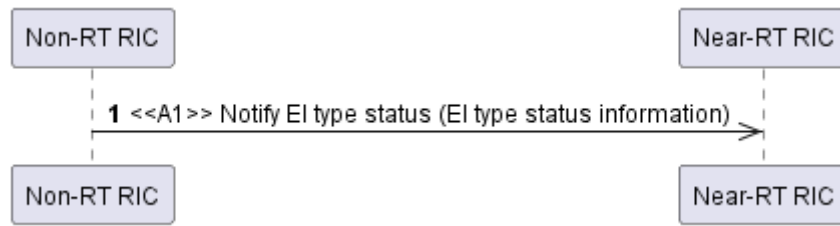


Figure 7.2.3.3-1: Notify EI type status

7.2.4 Required data

The EI type status information includes EI type availability change information (i.e. if EI type has been made available or unavailable) and/or state change information related to an EI type.

The state of the EI type state indicates if EI jobs can be created for the EI type or not, and if EI jobs for the EI type would be enabled or disabled.

The EI type status subscription information includes details on if notifications are requested for changes in EI type availability and/or changes in state of the EI type for which EI type identifiers were provided.

7.3 Create EI job use cases

7.3.1 Background and goal of the use cases

The create EI job use cases define how Near-RT RIC can create an EI job for an EI type and subscribe to notifications for changes in EI job status.

Near-RT RIC provides EI job definition information based on EI type information and considering the EI job constraints.

7.3.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:
 - a) Handles EI jobs for available EI types.
- 2) Near-RT RIC:
 - b) Creates EI job in Non-RT RIC.

7.3.3 Solutions

7.3.3.1 Create EI job

Table 7.3.3.1-1: Create EI job use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC to create EI job for an EI type	
Actors and Roles	Non-RT RIC as A1-EI Producer Near-RT RIC as A1-EI Consumer	
Assumptions	The EI type information known to the Near-RT RIC corresponds to an available EI type. Near-RT RIC has the schema for formulating EI job definition.	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-EI service. The EI type information is known to the Near-RT RIC.	
Begins when	Near-RT RIC initiates EI job creation	
Step 1 (M)	Near-RT RIC sends Create EI job request containing the EI job identifier and the EI job definition	
Step 2 (M)	Non-RT RIC sends Create EI job response	
Ends when	EI job has been created	
Exceptions		
Post-conditions	EI job exists	
Traceability	REQ-A1-EI-FUN6	

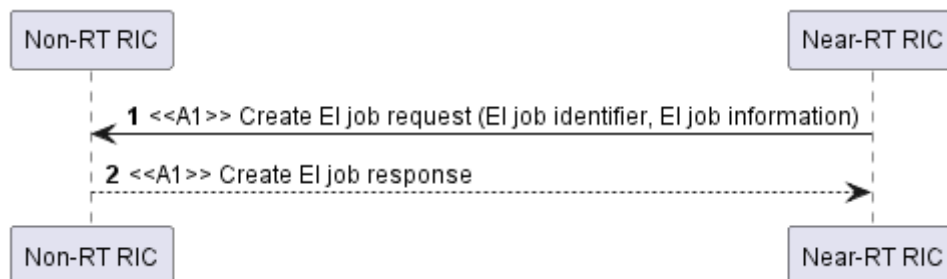


Figure 7.3.3.3-1: Create EI job

7.3.4 Required data

For creating an EI job of a certain EI type, the Near-RT RIC provides the EI job identifier and the EI job information, the callback URI for EI job result delivery, and optionally the callback URI for EI job status notifications.

EI job information includes the EI job definition that is formulated based on, and validated against, the EI job definition schema. The EI job definition can be formulated considering the EI job constraints information if provided.

7.4 Query EI jobs use cases

7.4.1 Background and goal of the use cases

The query EI jobs use cases define how Near-RT RIC can query for information on existing EI jobs.

7.4.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:

- a) Handles EI job.
- b) Responds to queries for EI job identifiers and EI job information.
- 2) Near-RT RIC:
 - a) Discovers EI jobs available in Non-RT RIC.
 - b) Retrieves EI job information.

7.4.3 Solutions

7.4.3.1 Query EI job identifiers

Table 7.4.3.1-1: Query EI job identifiers use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC to discover job identifiers for EI jobs that exist in Non-RT RIC	
Actors and Roles	Non-RT RIC as A1-EI Producer Near-RT RIC as A1-EI Consumer	
Assumptions	EI jobs exist in Non-RT RIC	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-EI service	
Begins when	Near-RT RIC initiates EI job identifiers query	
Step 1 (M)	Near-RT RIC sends Query EI job identifiers request	
Step 2 (M)	Non-RT RIC sends Query EI job identifiers response containing EI job identifiers	
Ends when	Near-RT RIC has received EI job identifiers	
Exceptions		
Post-conditions	EI job identifiers are known to Non-RT RIC	
Traceability	REQ-A1-EI-FUN7	

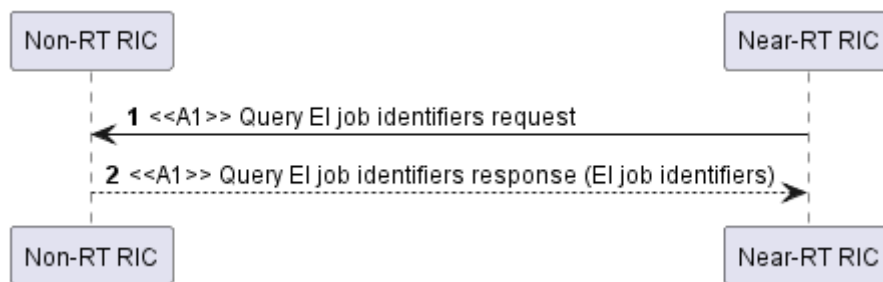


Figure 7.4.3.1-1: Query EI job identifiers

7.4.3.2 Query EI job

Table 7.4.3.2-1: Query EI job use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC to retrieve information about an EI job	
Actors and Roles	Non-RT RIC as A1-EI Producer Near-RT RIC as A1-EI Consumer	
Assumptions	The EI job identifier known to the Near-RT RIC corresponds to an existing EI job. Near-RT RIC has the schema for interpreting EI job information.	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-EI service. The EI job identifier is known to the Near-RT RIC.	
Begins when	Near-RT RIC initiates EI job query	
Step 1 (M)	Near-RT RIC sends Query EI job request containing the EI job identifier	
Step 2 (M)	Non-RT RIC sends Query EI job response containing EI job information	
Ends when	Near-RT RIC has received the EI job information	
Exceptions		
Post-conditions	EI job information is known to Near-RT RIC	
Traceability	REQ-A1-EI-FUN8	

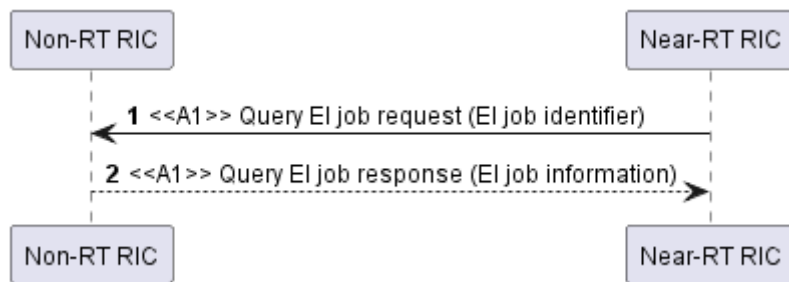


Figure 7.4.3.3-1: Query EI job

7.4.4 Required data

For querying EI job identifiers, the Near-RT RIC provides optionally an EI type identifier as filter parameter.

For querying EI job, the Near-RT RIC provides an EI job identifier.

EI job information includes the EI job definition.

7.5 Update EI job use cases

7.5.1 Background and goal of the use cases

The update EI job use cases define how Near-RT RIC can update an existing EI job.

Near-RT RIC provides updated EI job definition information. The reason for the update can be related to state of the EI type, EI job status and previously received EI job results.

7.5.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:

- a) Handles EI jobs for available EI types.
- 2) Near-RT RIC:
 - b) Updates EI job in Non-RT RIC that it has created.

7.5.3 Solutions

7.5.3.1 Update EI job

Table 7.5.3.1-1: Update EI job use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC to update an existing EI job	
Actors and Roles	Non-RT RIC as A1-EI Producer Near-RT RIC as A1-EI Consumer	
Assumptions	The EI job identifier known to the Near-RT RIC corresponds to an existing EI job that was created by the Near-RT RIC	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-EI service. The EI job exists.	
Begins when	Near-RT RIC initiates EI job update	
Step 1 (M)	Near-RT RIC sends Update EI job request containing the updated EI job information	
Step 2 (M)	Non-RT RIC sends Update EI job response	
Ends when	EI job has been updated	
Exceptions		
Post-conditions	EI job exists	
Traceability	REQ-A1-EI-FUN6	

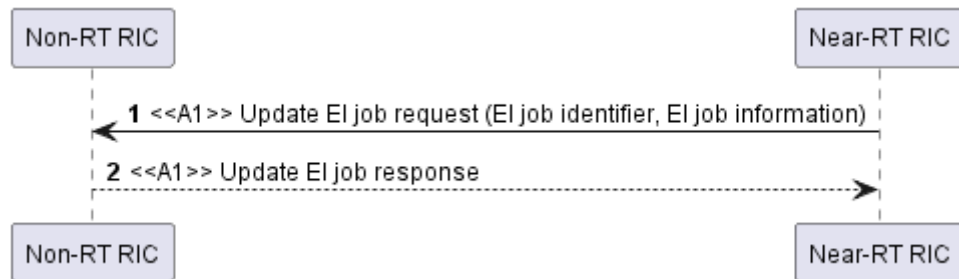


Figure 7.5.3.3-1: Update EI job

7.5.4 Required data

For updating an EI job, the Near-RT RIC provides the EI job identifier and updated EI job information and/or callback URI for EI job status notifications.

7.6 Delete EI job use cases

7.6.1 Background and goal of the use cases

The delete EI job use cases define how Near-RT RIC can delete an EI job.

The reason for the deletion can be related to the state of the EI type, EI job status and previously received EI job results.

7.6.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:
 - a) Handles EI jobs for available EI types.
- 2) Near-RT RIC:
 - b) Deletes EI job in Non-RT RIC that it has created.

7.6.3 Solutions

7.6.3.1 Delete EI job

Table 7.6.3.1-1: Delete EI job use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC to delete an existing EI job	
Actors and Roles	Non-RT RIC as A1-EI Producer Near-RT RIC as A1-EI Consumer	
Assumptions	The EI job identifier known to the Near-RT RIC corresponds to an existing EI job that was created by the Near-RT RIC	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-EI service. The EI job exists.	
Begins when	Near-RT RIC initiates EI job deletion	
Step 1 (M)	Near-RT RIC sends Delete EI job request containing the EI job identifier	
Step 2 (M)	Non-RT RIC sends Delete EI job response	
Ends when	EI job has been deleted	
Exceptions		
Post-conditions	EI job does not exist	
Traceability	REQ-A1-EI-FUN6	



Figure 7.6.3.3-1: Delete EI job

7.6.4 Required data

For deleting an EI job, the Near-RT RIC provides the EI job identifier.

7.7 EI job status use cases

7.7.1 Background and goal of the use cases

The EI job status use cases define how Near-RT RIC can detect status, and changes in status, of an EI job.

EI job status information is provided by the Non-RT RIC and is used by Near-RT RIC when managing EI jobs.

7.7.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:
 - a) Handles EI job and responds to queries for EI job status.
 - b) Notifies Near-RT RIC about changes in EI job status.
- 2) Near-RT RIC:
 - a) Requests EI job status.
 - b) Receives notifications about changes in EI job status.

7.7.3 Solutions

7.7.3.1 Query EI job status

Table 7.7.3.1-1: Query EI job status use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC to retrieve EI job status information for an existing EI job	
Actors and Roles	Non-RT RIC as A1-EI Producer Near-RT RIC as A1-EI Consumer	
Assumptions	The EI job identifier known to the Near-RT RIC corresponds to an existing EI job that was created by the Near-RT RIC. Near-RT RIC has the schema for interpreting EI job status information.	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-EI service. The EI job identifier is known to the Near-RT RIC.	
Begins when	Near-RT RIC initiates EI job status query	
Step 1 (M)	Near-RT RIC sends Query EI job status request containing the EI job identifier of the EI job being queried for status	
Step 2 (M)	Non-RT RIC sends Query EI job status response containing EI job status information	
Ends when	Near-RT RIC has received the EI job status information	
Exceptions		
Post-conditions	EI job status information is known to Near-RT RIC	
Traceability	REQ-A1-EI-FUN9	

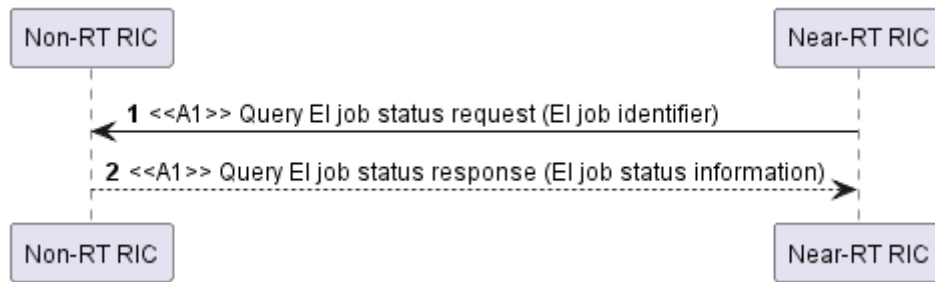


Figure 7.7.3.1-1: Query EI job status

7.7.3.2 Notify EI job status

Table 7.7.3.2-1: Notify EI job status use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC to receive EI job status information for an existing EI job	
Actors and Roles	Non-RT RIC as A1-EI Producer Near-RT RIC as A1-EI Consumer	
Assumptions	Near-RT RIC has the schema for interpreting EI job status information.	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-EI service. A callback URI for EI job status notifications has been provided when creating and/or updating the EI job.	
Begins when	Event occurs in Non-RT RIC related to the status of the EI job	
Step 1 (M)	Non-RT RIC sends Notify EI job status request containing the EI job status information	
Ends when	Near-RT RIC has received the EI job status information	
Exceptions		
Post-conditions	EI job status information is known to Near-RT RIC	
Traceability	REQ-A1-EI-FUN10	



Figure 7.7.3.2-1: Notify EI job status

7.7.4 Required data

EI job status information includes the EI job status that is formulated based on, and validated against, the EI job status schema. The EI job status information includes indication if EI job is enabled or disabled.

7.8 EI delivery use cases

7.8.1 Background and goal of the use cases

The EI delivery use cases define how Non-RT RIC delivers EI job results to Near-RT RIC based on an EI job.

Depending on the EI job definition, the EI job result can be delivered in one delivery message or in repeated delivery messages.

7.8.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:
 - a) Handles EI job and delivers EI job results.
- 2) Near-RT RIC:
 - b) Receives EI job result related to an EI job it has created.

7.8.3 Solutions

7.8.3.1 Deliver EI job result

Table 7.8.3.1-1: Deliver EI job result use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to deliver EI job result based on an EI job	
Actors and Roles	Non-RT RIC as A1-EI Producer Near-RT RIC as A1-P Consumer	
Assumptions	There is enrichment information based on the EI job definition that is to be delivered	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-EI service. EI job exists and is enabled.	
Begins when	Non-RT RIC initiates EI job result delivery	
Step 1 (M)	Non-RT RIC sends push request with EI job result to be delivered	
Step 2 (M)	Near-RT RIC validates the received information and sends push response	
Ends when	Near-RT RIC has received enrichment information	
Exceptions		
Post-conditions	EI job result for the EI job has been delivered	
Traceability	REQ-A1-EI-FUN11	

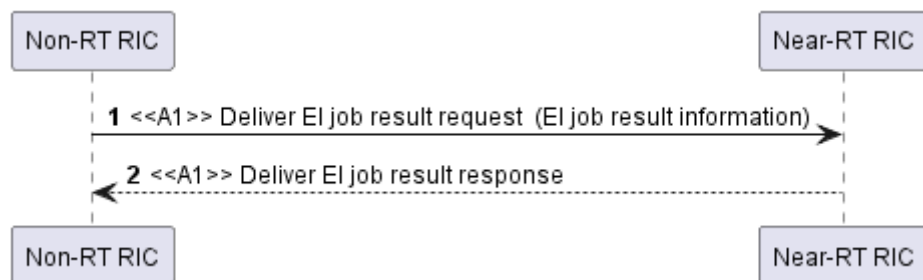


Figure 7.8.3.1-1: Deliver EI job result

7.8.3.2 Deliver EI job results

Table 7.8.3.2-1: Deliver EI job results use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to deliver EI job results based on an EI job	
Actors and Roles	Non-RT RIC as A1-EI Producer Near-RT RIC as A1-P Consumer	
Assumptions	There is enrichment information based on the EI job definition that is to be delivered in more than one delivery message	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-EI service. EI job exists and is enabled.	
Begins when	Non-RT RIC initiates EI job result delivery	
Step 1 – ref (M)	Non-RT RIC delivers EI job result	7.8.3.1
Step 2 – loop (M)	Non-RT RIC repeats Step 1 as per the EI job definition	
Ends when	The EI job is completed or deleted.	
Exceptions		
Post-conditions	EI job result for the EI job has been delivered	
Traceability	REQ-A1-EI-FUN11	

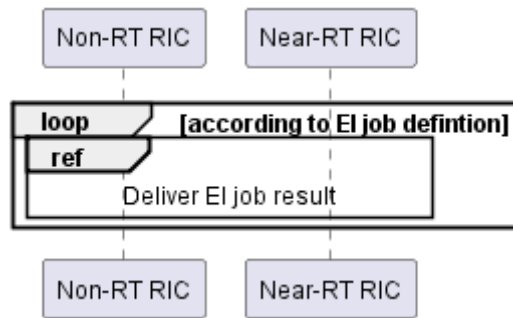


Figure 7.8.3.2-1: Deliver EI job results

7.8.4 Required data

EI job result information contains the enrichment information requested in the EI job definition. It is formulated based on, and validated against, the EI job result schema.

8 Use cases for A1 ML model management service – Near-RT RIC as A1-ML Consumer

8.1 Request AI/ML model training use cases

8.1.1 Background and goal of the use cases

The Request AI/ML model training use cases define how Near-RT RIC requests SMO/Non-RT RIC to train an AI/ML Model.

8.1.2 Entities/resources involved in the use cases

- 1) SMO/Non-RT RIC:

- a) Produces AI/ML model training services.
- b) Makes AI/ML model training services available to Near-RT RIC.
- 2) Near-RT RIC:
 - b) Requests AI/ML model training from SMO/Non-RT RIC.

8.1.3 Solutions

8.1.3.1 Request AI/ML model training

Table 8.1.3.1-1: Request AI/ML model training use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC requests training of an AI/ML model.	
Actors and Roles	SMO/Non-RT RIC as producer of AI/ML training services. Non-RT RIC as A1-ML Producer. Near-RT RIC as A1-ML Consumer.	
Assumptions	AI/ML training services are produced by the SMO/Non-RT RIC framework or by rApps for the AI/ML model that training is requested for. Near-RT RIC is aware of that SMO/Non-RT RIC can train the AI/ML model.	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-ML service.	
Begins when	Near-RT RIC determines the need to train an AI/ML model	
Step 1 (M)	Near-RT RIC sends Request training request	
Step 2 (M)	Non-RT RIC sends Request training response	
Step 3 (M)	SMO/Non-RT RIC trains the AI/ML model	
Step 4 (M)	Non-RT RIC sends Notify AI/ML model training job status message containing the AI/ML model training job status and status-dependent information	
Ends when	Near-RT RIC has the location of the trained AI/ML model	
Exceptions		
Post-conditions	Trained AI/ML model is available to Near-RT RIC	
Traceability	REQ-A1-ML-FUN2	

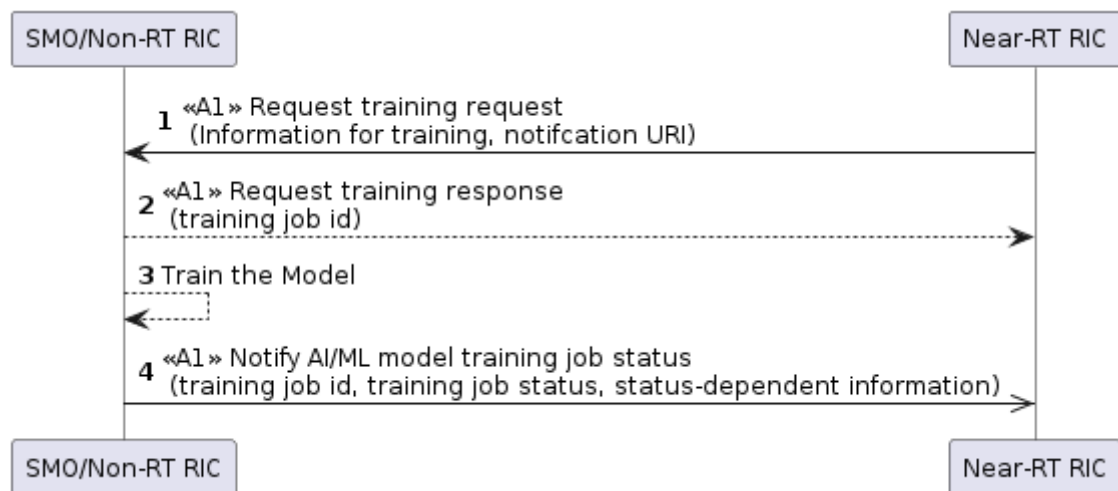


Figure 8.1.3.1-1: Request AI/ML model training - Near-RT RIC requests AI/ML model training

8.1.4 Required data

Information for training (e.g. information about training dataset, model identifier, training criteria), and a notification URI to receive notification is provided by Near-RT RIC.

SMO/Non-RT RIC provides the training job identifier and the information for training job status notification, including status-dependent information (e.g location of the trained model, or details in case of non-successful training).

8.2 AI/ML model training job status use cases

8.2.1 Background and goal of the use cases

This use cases define how Near-RT RIC queries AI/ML model training job status and receives notification about AI/ML model training job status change. AI/ML model training job status is provided by the SMO/Non-RT RIC.

8.2.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:
 - a) Responds to queries for AI/ML model training job status.
 - b) Notifies Near-RT RIC about changes in AI/ML model training job status.
- 2) Near-RT RIC:
 - a) Requests AI/ML model training job status.
 - b) Receives notifications about changes in AI/ML model training job status.

8.2.3 Solutions

8.2.3.1 Query AI/ML model training job status

Table 8.2.3.1-1: Query AI/ML model training job status use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC to retrieve AI/ML model training job status for an existing AI/ML model training job	
Actors and Roles	Non-RT RIC as A1-ML Producer Near-RT RIC as A1-ML Consumer	
Assumptions	The AI/ML model training job identifier known to the Near-RT RIC corresponds to an existing AI/ML model training job that was created based on the training request from Near-RT RIC.	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-ML service. The AI/ML model training job identifier is known to the Near-RT RIC.	
Begins when	Near-RT RIC initiates AI/ML model training job status query	
Step 1 (M)	Near-RT RIC sends Query AI/ML model training job status request containing the AI/ML model training job identifier of the AI/ML model training job being queried for status	
Step 2 (M)	Non-RT RIC sends Query AI/ML model training job status response containing AI/ML model training job status	
Ends when	Near-RT RIC has received the AI/ML model training job status	
Exceptions		
Post-conditions	AI/ML model training job status is known to Near-RT RIC	
Traceability	REQ-A1-ML-FUN3	

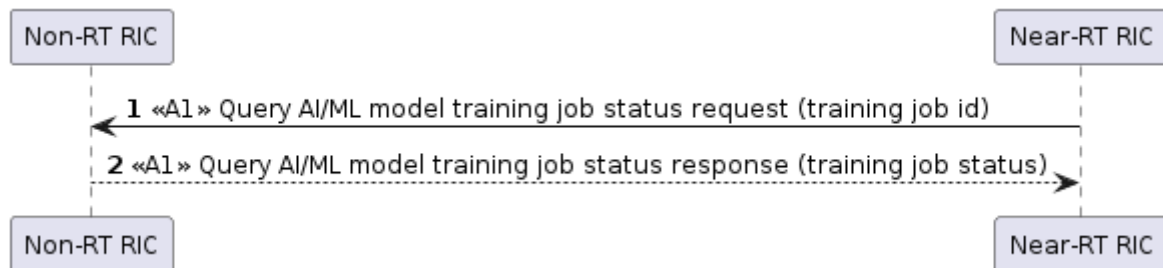


Figure 8.2.3.1-1: Query AI/ML model training job status

8.2.3.2 Notify AI/ML model training job status

Table 8.2.3.2-1: Notify AI/ML model training job status use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC to receive AI/ML model training job status for an existing AI/ML model training job	
Actors and Roles	Non-RT RIC as A1-ML Producer Near-RT RIC as A1-ML Consumer	
Assumptions		
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-ML service. A callback URI for AI/ML model training job status notifications has been provided when creating the AI/ML model training job.	
Begins when	Event occurs in SMO/Non-RT RIC related to the status of the AI/ML model training job	
Step 1 (M)	Non-RT RIC sends Notify AI/ML model training job status message containing the AI/ML model training job status and optionally status-dependent information	
Ends when	Near-RT RIC has received the AI/ML model training job status	
Exceptions		
Post-conditions	AI/ML model training job status is known to Near-RT RIC	
Traceability	REQ-A1-ML-FUN4	

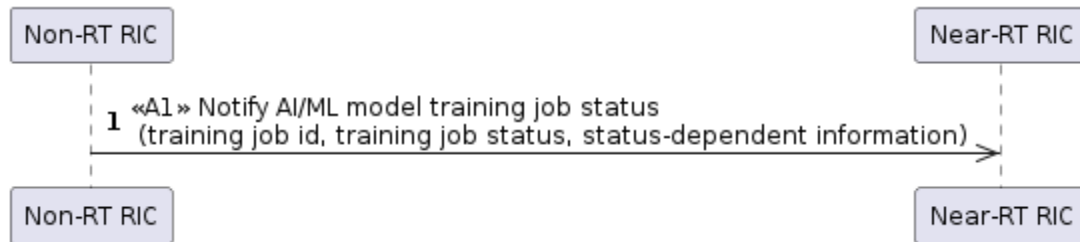


Figure 8.2.3.2-1: Notify AI/ML model training job status

8.2.4 Required data

SMO/Non-RT RIC provides the training job id, the training job status, and optionally the status-dependent information.

8.3 Discover AI/ML model training capabilities use cases

8.3.1 Background and goal of the use cases

These use cases define how Near-RT RIC discovers AI/ML model training capabilities. AI/ML model training capabilities are provided by the Non-RT RIC.

8.3.2 Entities/resources involved in the use cases

- 1) SMO/Non-RT RIC:
 - a) Provides information on AI/ML model training capabilities.
 - b) Makes AI/ML model training capabilities available to Near-RT RIC.
- 2) Near-RT RIC:

- a) Discovers AI/ML model training capabilities.

8.3.3 Solutions

8.3.3.1 Discover AI/ML model training capabilities

Table 8.3.3.1-1 Discover AI/ML model training capabilities use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Near-RT RIC is to discover AI/ML model training capabilities provided by Non-RT RIC.	
Actors and Roles	Non-RT RIC as A1-ML Producer Near-RT RIC as A1-ML Consumer	
Assumptions	AI/M model training capabilities are provided by the SMO/Non-RT RIC framework or by rApps that registered the AI/ML model training capabilities in the Non-RT RIC.	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-ML service.	
Begins when	Near-RT RIC determines the need to discover AI/ML model training capabilities provided by Non-RT RIC	
Step 1 (M)	Near-RT RIC sends Query AI/ML model training capabilities request for the information on the AI/ML model training capabilities by providing selection criteria	
Step 2 (M)	Non-RT RIC looks up the information of the discovered AI/ML model training capabilities	
Step 3 (M)	Non-RT RIC sends Query AI/ML model training capabilities response containing the information of AI/ML model training capabilities	
Ends when	Near-RT RIC has received the AI/ML model training capabilities	
Exceptions		
Post-conditions	AI/ML model training capabilities are known to Near-RT RIC	
Traceability	REQ-A1-ML-FUN5	

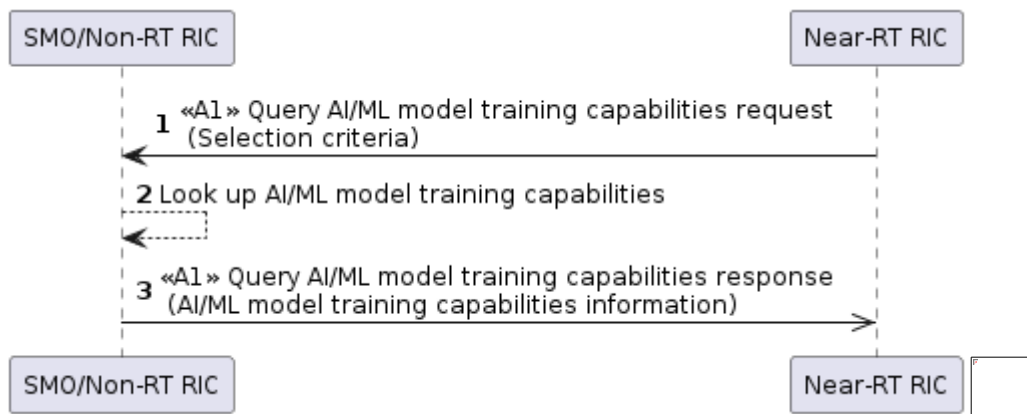


Figure 8.3.3.1-1 Discover AI/ML model training capabilities

8.3.4 Required data

The selection criteria for discovery AI/ML model training capabilities is provided by Near-RT RIC.

SMO/Non-RT RIC provides the AI/ML model training capabilities.

9 Use cases for A1 ML model management service – Non-RT RIC as A1-ML Consumer

9.1 Request AI/ML model training use cases

9.1.1 Background and goal of the use cases

The Request AI/ML model training use cases define how Non-RT RIC requests Near-RT RIC to train an AI/ML Model.

9.1.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:
 - a) Requests AI/ML model training from Non-RT RIC.
- 2) Near-RT RIC:
 - a) Produce AI/ML model training services.
 - b) Makes AI/ML model training services available to Non-RT RIC.

9.1.3 Solutions

9.1.3.1 Request AI/ML model training

Table 9.1.3.1-1: Request AI/ML model training use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC requests training of an AI/ML model.	
Actors and Roles	Near-RT RIC as producer of AI/ML training services and A1-ML Producer. Non-RT RIC as A1-ML Consumer.	
Assumptions	Non-RT RIC is aware of that Near-RT RIC can train the AI/ML model.	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-ML service.	
Begins when	Non-RT RIC determines the need to train an AI/ML model	
Step 1 (M)	Non-RT RIC sends Request training request	
Step 2 (M)	Near-RT RIC sends Request training response	
Step 3 (M)	Near-RT RIC trains the AI/ML model	
Step 4 (M)	Near-RT RIC sends Notify AI/ML model training job status message containing the AI/ML model training job status and status-dependent information	
Ends when	Non-RT RIC has the location of the trained AI/ML model	
Exceptions		
Post-conditions	Trained AI/ML model is available to Non-RT RIC	
Traceability	REQ-A1-ML-FUN2	

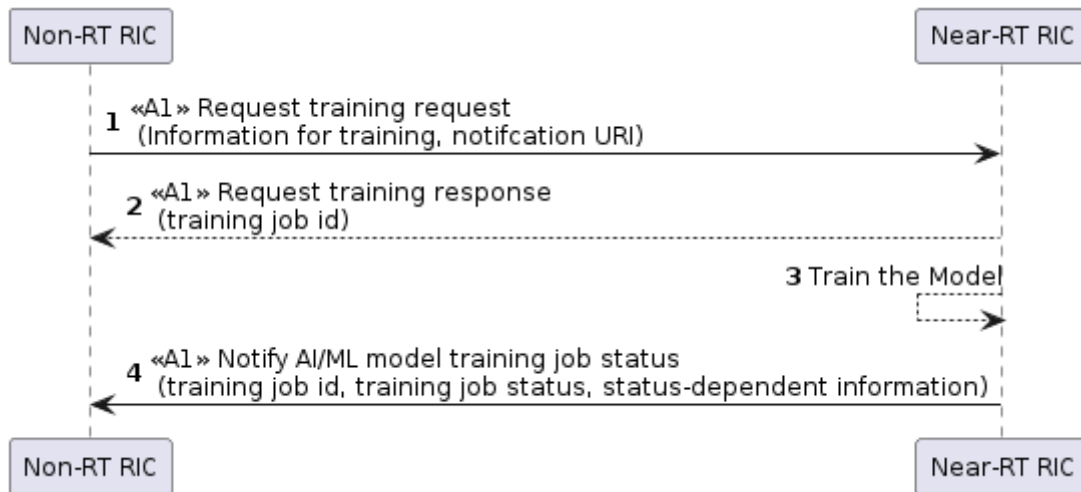


Figure 9.1.3.1-1: AI/ML model training - Non-RT RIC requests AI/ML model training

9.1.4 Required data

Information for training (e.g. information about training dataset, model identifier, training criteria), and a notification URI to receive notification is provided by Non-RT RIC.

Near-RT RIC provides the training job identifier and the information for training job status notification, including status-dependent information (e.g location of the trained model, or details in case of non-successful training).

9.2 AI/ML model training job status use cases

9.2.1 Background and goal of the use cases

These use cases define how Non-RT RIC queries AI/ML model training job status and receives notification about AI/ML model training job status change. AI/ML model training job status is provided by the Near-RT RIC.

9.2.2 Entities/resources involved in the use cases

- 1) Non-RT RIC:
 - a) Requests AI/ML model training job status.
 - b) Receives notifications about changes in AI/ML model training job status.
- 2) Near-RT RIC:
 - a) Responds to queries for AI/ML model training job status.
 - b) Notifies Non-RT RIC about changes in AI/ML model training job status.

9.2.3 Solutions

9.2.3.1 Query AI/ML model training job status

Table 9.2.3.1-1: Query AI/ML model training job status use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to retrieve AI/ML model training job status for an existing AI/ML model training job	
Actors and Roles	Near-RT RIC as A1-ML Producer Non-RT RIC as A1-ML Consumer	
Assumptions	The AI/ML model training job identifier known to the Non-RT RIC corresponds to an existing AI/ML model training job that was created based on the training request from Non-RT RIC.	
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-ML service. The AI/ML model training job identifier is known to the Non-RT RIC.	
Begins when	Non-RT RIC initiates AI/ML model training job status query	
Step 1 (M)	Non-RT RIC sends Query AI/ML model training job status request containing the AI/ML model training job identifier of the AI/ML model training job being queried for status	
Step 2 (M)	Near-RT RIC sends Query AI/ML model training job status response containing AI/ML model training job status	
Ends when	Non-RT RIC has received the AI/ML model training job status	
Exceptions		
Post-conditions	AI/ML model training job status is known to Non-RT RIC	
Traceability	REQ-A1-ML-FUN3	

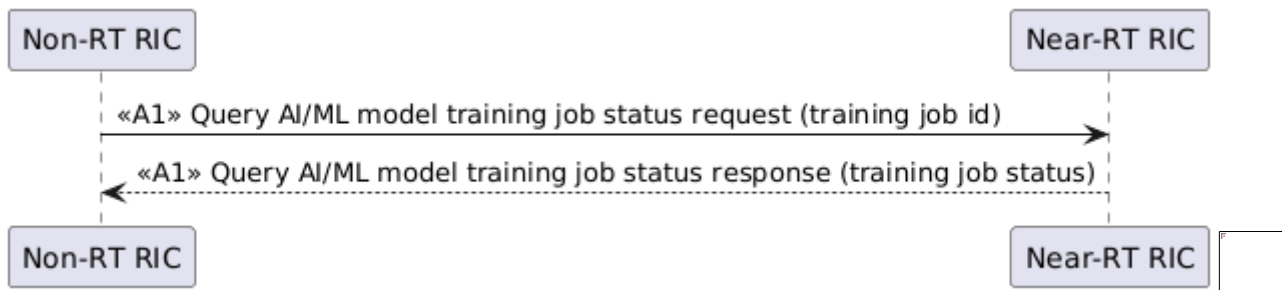


Figure 9.2.3.1-1: Query AI/ML model training job status

9.2.3.2 Notify AI/ML model training job status

Table 9.2.3.2-1: Notify AI/ML model training job status use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	Non-RT RIC to receive AI/ML model training job status for an existing AI/ML model training job	
Actors and Roles	Near-RT RIC as A1-ML Producer Non-RT RIC as A1-ML Consumer	
Assumptions		
Pre-conditions	A1 interface is established, and the actors are authorized for using the A1-ML service. A callback URI for AI/ML model training job status notifications has been provided when creating the AI/ML model training job.	
Begins when	Event occurs in Near-RT RIC related to the status of the AI/ML model training job	
Step 1 (M)	Near-RT RIC sends Notify AI/ML model training job status message containing the AI/ML model training job status and optionally status-dependent information	
Ends when	Non-RT RIC has received the AI/ML model training job status	
Exceptions		
Post-conditions	AI/ML model training job status is known to Non-RT RIC	
Traceability	REQ-A1-ML-FUN4	

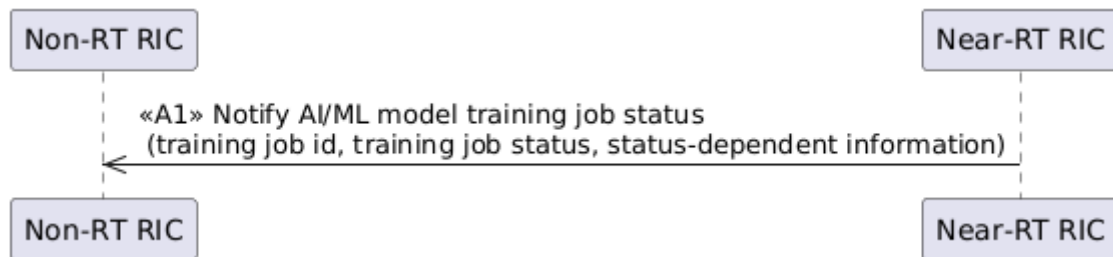


Figure 9.2.3.2-1: Notify AI/ML model training job status

9.2.4 Required data

Near-RT RIC provides the training job id, the training job status, and optionally the status-dependent information (e.g location of the trained model, or details in case of non-successful training).

Annex A (normative): Authorization of service access requests

A.1 Pre-conditions

The preconditions of the A1 use cases in clauses 6 to 8 are that the actors are authorized for using the A1 service and the authorization procedures are defined in SPS [9], clause 4.7. The registration process for the entity acting as service producer is described in SPS [9], clause 4.7.2.2, and the access token request process for the entity acting as service consumer is described in SPS [9], clause 4.7.2.3.

A.2 A1 service access based on token verification use cases

A.2.1 Background and goal of the use cases

The service access request based on token verification is described generically in SPS [9], clause 4.7.2.4.

The A1 service access use cases define how the access token is transferred and verified. This description applies to all solutions specified in clauses 6 to 8 although they do not explicitly refer to token transfer or verification.

A.2.2 Entities/resources involved in the use cases

- 1) SMO/Non-RT RIC framework:
 - a) either makes service access requests containing access token; or
 - b) verifies received access token before sending service access responses.
- 2) Near-RT RIC:
 - a) either verifies received access token before sending service access responses; or
 - b) makes service access requests containing access token.

A.2.3 Solutions

A.2.3.1 General

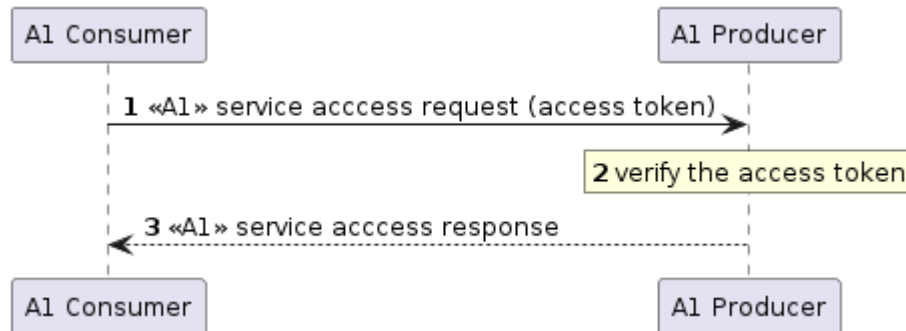
The notation A1 Consumer refers to either A1-P Consumer, A1-EI Consumer, or A1-ML Consumer while the notation A1 Producer refers to either A1-P Producer, A1-EI Producer, or A1-ML Producer. SMO/Non-RT RIC framework acts as A1 Consumer for A1-P and/or A1-ML when Near-RT RIC acts as A1 Producer for A1-P and/or A1-ML, and SMO/Non-RT RIC framework acts as A1 Producer for A1-EI and/or A1-ML when Near-RT RIC acts as A1 Consumer for A1-EI and/or A1-ML.

The notation “service access” for the use case, the procedure, and the request/response messages in this clause refers to the use case, procedure and request/response message names used in the solutions in clauses 6 to 8.

A.2.3.2 Service access request

Table A.2.3.2-1: Service access use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	A1 Producer to verify access token in service access request received from A1 Consumer	SPS [9], clause 4.7.2.4
Actors and Roles	SMO/Non-RT RIC framework as A1 Consumer and Near-RT RIC as A1 Producer or SMO/Non-RT RIC framework as A1 Producer and Near-RT RIC as A1 Consumer	
Assumptions	See solutions for A1-P, A1-EI, and A1-ML use cases	Clauses 6 to 8
Pre-conditions	A1 interface is established, and the actors are authorized for using the requested A1 service.	SPS [9], clause 4.7
Begins when	A1 Consumer initiates service access	
Step 1 (M)	A1 Consumer sends A1 service access request message containing access token	Clauses 6 to 8
Step 2 (M)	A1 Producer verifies the access token.	
Step 3 (M)	A1 Producer sends A1 service access response message.	Clauses 6 to 8
Ends when	Service access request has been verified.	
Exceptions	If token verification was not successful, further details may be provided that may indicate what is needed to access the service.	
Post-conditions	If verification was successful, the service has been executed.	
Traceability	REQ-A1-P-FUN11, REQ-A1-EI-FUN12, REQ-A1-ML-FUN1	


Figure A.2.3.2-1: Service access

A.2.4 Required data

In addition to the data required in the solutions in clauses 6 to 8, the access token is included in the service access request message.

In case service access is rejected due to that access is forbidden, the service access response message may include information on the token scope necessary to access the service.

Annex (informative): Change History

Date	Revision	Description
2022.07.30	01.00	First version
2022.11.17	01.01	Aligning to O-RAN drafting rules
2023.11.30	01.02	Editorial enhancement and applying latest template
2024.03.31	01.03	Editorial enhancements and alignment of notation for status and feedback.
2024.07.31	01.04	Updated specification designator to R004
2024.11.30	02.00	Use cases for Authorization of service access requests and AI/ML model training
2025.03.31	03.00	Use case for AI/ML model training job status with Non-RT RIC as A1-ML Consumer
2025.07.31	04.00	Use case for AI/ML model training capabilities discovery